PIPING & INSTRUMENTATION DIAGRAM (P&ID) — DISTILLATION SYSTEM (C-201)

ISA-5.1 INSTRUMENTATION LOOPS: REFLUX CASCADE, SPLIT-RANGE PRESSURE, SUMP LEVEL & STEAM INTERLOCKS

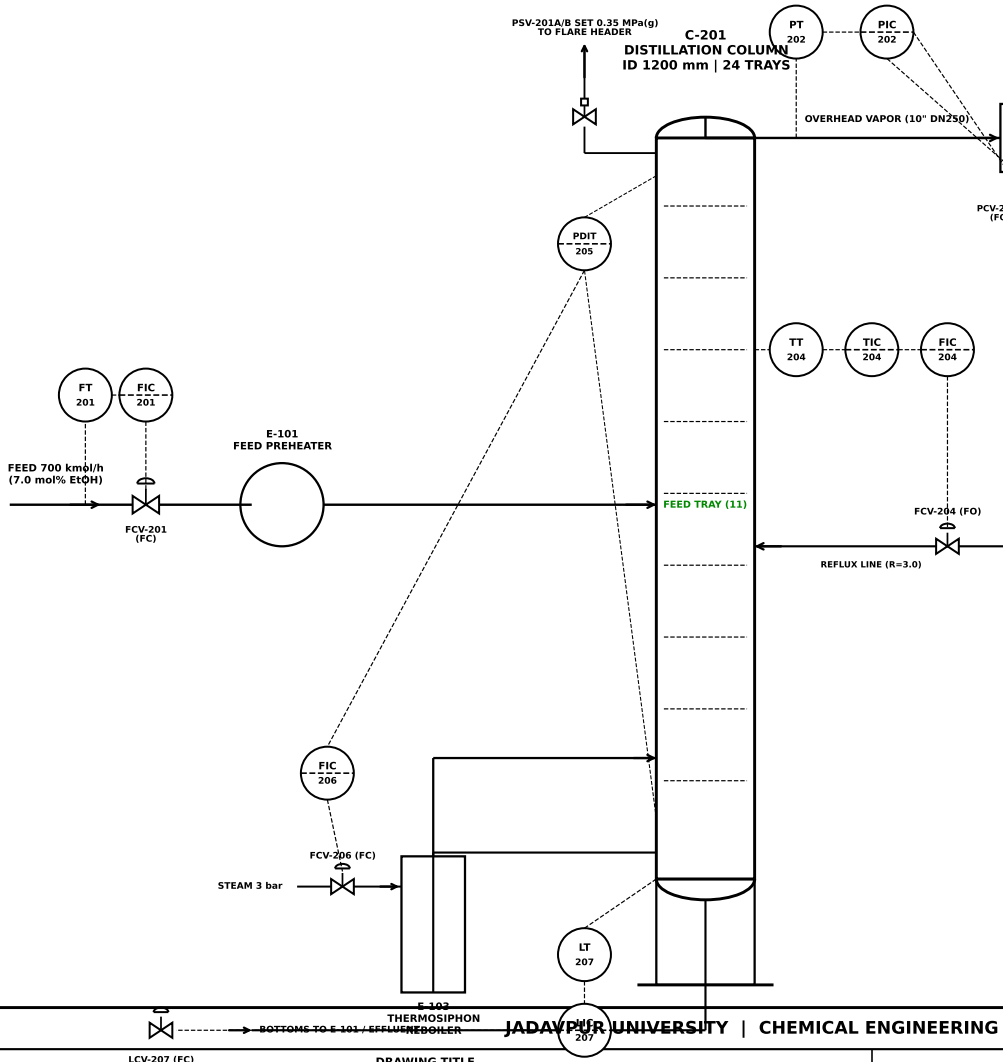

TABLE 1: ISA-5.1 INSTRUMENT & CONTROL LOOP SCHEDULE

TAG NO.	SERVICE / LOOP	RANGE	SETPOINT	ACTION / FAIL
FT/FIC-201	Feed Flow to Column	0 - 1000 kmol/h	700.0 kmol/h	FCV-201 (Fail Closed)
PT/PIC-202	Top Vapor Pressure	0 - 200 kPa(a)	101.325 kPa	Split: PCV-202A (FO) / B (FC)
LT/LIC-203	Reflux Drum Level	0 - 100 %	50.0 % level	LCV-203 (Fail Closed)
TT/TIC-204	Tray 18 Temperature	50 - 120 C	79.2 C	Cascades to FIC-204
FT/FIC-204	Reflux Liquid Flow	0 - 300 kmol/h	167.7 kmol/h	FCV-204 (Fail Open - FO)
PDIT-205	Column Differential Press.	0 - 30.0 kPa	13.0 kPa	Alarm High @ 17.5 kPa
FT/FIC-206	Reboiler 3 bar Steam	0 - 6,000 kg/h	4,850 kg/h	FCV-206 (Fail Closed)
LT/LIC-207	Bottoms Sump Level	0 - 100 %	50.0 % level	LCV-207 (Fail Closed)
PSV-201A/B	Column Overpressure Relief	0 - 0.50 MPa	0.35 MPa(g)	Pop Action to Flare Header

TABLE 2: SAFETY INTERLOCK & CAUSE-AND-EFFECT MATRIX

TRIP ID	INITIATING EVENT	LOGIC / SETPOINT	AUTOMATED SAFETY ACTION
I-01	Column High DP (Flooding)	PDIT-205 > 17.5 kPa	Cut reboiler steam FCV-206 to 30%; Alarm
I-02	Low Sump Level (Dryout)	LT-207 < 15.0 %	Trip steam FCV-206 closed; Stop bottoms draw
I-03	Reflux Pump Failure (No flow)	FIC-204 < 20.0 kmol/h	Auto-start standby P-102B; Hold feed rate
I-04	High Column Pressure (Overpress)	PT-202 > 135 kPa(a)	Open PCV-202B vent to flare; Trip steam
I-05	Reflux Drum High Level	LT-203 > 85.0 %	Open LCV-203 full; Cut column feed rate

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PIPING & INSTRUMENTATION DIAGRAM (P&ID) — ISA-5.1	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	SCHEMATIC (ISA-5.1)
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/ED/DWG-05	01 / 09 / 2026

ETHANOL DISTILLATION (C-201) | 700 kmol/h FEED (7 mol% EtOH) | ID 1.20 m, R=3.0 | ASME SEC VIII / TEMA